

## AURORA, IL, USA

WMO: 744655

Lat: 41.770N

Lon: 88.481W

Elev: 216

StdP: 98.75

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 04808

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -21.1      | -17.7 | -24.6                         | 0.4 | -20.7 | -21.8 | 0.5 | -17.1 | 12.6                  | -2.9 | 11.6 | -4.5 | 3.4                   | 250  | 0.558 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.8                   | 32.5            | 23.6 | 31.2 | 23.0 | 29.8 | 22.4 | 25.4                | 30.5 | 24.5 | 29.1 | 23.6 | 28.0 | 4.9                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.9                            | 19.2 | 28.6 | 22.9 | 18.1 | 27.4 | 22.2 | 17.3 | 26.6 | 79.0          | 30.4 | 75.0 | 29.0 | 71.4 | 28.1 | 29.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 11.6              | 10.3 | 8.9 | -26.5                      | 34.6 | 3.9                | 1.8 | -29.3                                              | 35.9 | -31.6      | 36.9 | -33.8      | 37.9 | -36.6      | 39.2 |    |    |
|                   |      |     | -26.7                      | 27.1 | 3.8                | 0.8 | -29.4                                              | 27.7 | -31.6      | 28.2 | -33.7      | 28.6 | -36.5      | 29.2 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 9.6    | -5.2  | -3.7  | 3.1   | 9.5   | 15.7  | 20.9  | 22.7  | 21.4  | 17.8  | 11.0  | 4.0   | -2.3 |
|                                                                                  | DBStd       | 11.11  | 6.80  | 6.68  | 6.01  | 5.07  | 4.91  | 3.72  | 3.19  | 2.94  | 4.42  | 5.21  | 5.37  | 6.59 |
|                                                                                  | HDD10.0     | 1789   | 471   | 386   | 228   | 69    | 7     | 0     | 0     | 0     | 2     | 52    | 190   | 384  |
|                                                                                  | HDD18.3     | 3594   | 729   | 618   | 473   | 267   | 113   | 17    | 3     | 7     | 62    | 234   | 429   | 641  |
|                                                                                  | CDD10.0     | 1656   | 0     | 1     | 14    | 54    | 184   | 326   | 393   | 352   | 237   | 82    | 11    | 2    |
|                                                                                  | CDD18.3     | 421    | 0     | 0     | 1     | 3     | 31    | 93    | 138   | 101   | 47    | 6     | 0     | 0    |
|                                                                                  | CDH23.3     | 4129   | 0     | 0     | 9     | 48    | 325   | 913   | 1386  | 909   | 471   | 68    | 0     | 0    |
|                                                                                  | CDH26.7     | 1334   | 0     | 0     | 1     | 7     | 94    | 318   | 494   | 264   | 143   | 13    | 0     | 0    |
| Wind                                                                             | WSAvg       | 4.1    | 4.7   | 4.8   | 4.8   | 5.1   | 4.5   | 3.7   | 2.9   | 2.6   | 3.0   | 3.8   | 4.4   | 4.5  |
| Precipitation                                                                    | PrecAvg     | 905    | 38    | 38    | 59    | 87    | 108   | 102   | 98    | 105   | 81    | 72    | 61    | 50   |
|                                                                                  | PrecMax     | 1162   | 89    | 76    | 119   | 180   | 180   | 216   | 168   | 255   | 347   | 177   | 137   | 125  |
|                                                                                  | PrecMin     | 557    | 13    | 6     | 21    | 35    | 41    | 20    | 48    | 45    | 13    | 16    | 17    | 17   |
|                                                                                  | PrecStd     | 154    | 22    | 20    | 29    | 36    | 39    | 52    | 41    | 48    | 78    | 41    | 30    | 31   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 12.8  | 16.8  | 24.5  | 27.6  | 31.8  | 33.7  | 35.1  | 32.9  | 32.4  | 28.6  | 21.1  | 16.0 |
|                                                                                  |             | MCWB   | 10.5  | 12.6  | 17.7  | 17.4  | 21.0  | 23.1  | 25.3  | 24.3  | 23.3  | 21.0  | 14.5  | 14.2 |
|                                                                                  | 2%          | DB     | 8.5   | 11.8  | 18.8  | 24.3  | 29.1  | 32.2  | 32.6  | 31.2  | 30.6  | 25.1  | 17.5  | 12.3 |
|                                                                                  |             | MCWB   | 6.9   | 8.9   | 13.7  | 16.0  | 19.7  | 22.6  | 24.6  | 23.7  | 22.2  | 17.6  | 13.0  | 10.1 |
|                                                                                  | 5%          | DB     | 5.6   | 7.5   | 15.9  | 22.1  | 27.2  | 30.4  | 31.1  | 29.4  | 28.0  | 22.6  | 15.2  | 8.7  |
|                                                                                  |             | MCWB   | 4.1   | 5.4   | 11.7  | 14.5  | 18.8  | 21.9  | 23.8  | 22.8  | 20.4  | 16.7  | 11.9  | 7.3  |
|                                                                                  | 10%         | DB     | 3.3   | 4.5   | 12.5  | 19.0  | 24.5  | 28.2  | 29.3  | 27.9  | 26.2  | 19.9  | 12.6  | 6.3  |
|                                                                                  |             | MCWB   | 2.1   | 2.8   | 8.9   | 13.0  | 17.9  | 20.9  | 22.7  | 21.7  | 19.6  | 15.5  | 9.5   | 4.5  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 11.6  | 13.5  | 17.7  | 19.3  | 23.2  | 25.6  | 27.1  | 25.9  | 24.8  | 22.0  | 16.3  | 14.4 |
|                                                                                  |             | MCDB   | 12.3  | 15.5  | 23.8  | 25.2  | 29.1  | 30.8  | 32.1  | 31.1  | 30.3  | 27.4  | 18.9  | 15.7 |
|                                                                                  | 2%          | WB     | 7.0   | 9.1   | 14.9  | 17.2  | 21.4  | 24.1  | 25.9  | 24.8  | 23.4  | 19.6  | 14.1  | 10.4 |
|                                                                                  |             | MCDB   | 8.5   | 11.6  | 18.2  | 22.1  | 26.8  | 29.1  | 30.9  | 29.4  | 28.2  | 22.4  | 16.4  | 11.5 |
|                                                                                  | 5%          | WB     | 4.1   | 5.6   | 12.4  | 15.5  | 20.2  | 23.1  | 24.9  | 23.8  | 22.0  | 17.8  | 12.0  | 7.3  |
|                                                                                  |             | MCDB   | 5.2   | 7.7   | 15.0  | 20.8  | 25.1  | 28.3  | 29.7  | 27.9  | 26.0  | 21.1  | 14.5  | 8.5  |
|                                                                                  | 10%         | WB     | 2.2   | 2.9   | 9.1   | 13.8  | 18.9  | 22.1  | 24.0  | 22.8  | 20.8  | 16.0  | 10.0  | 4.6  |
|                                                                                  |             | MCDB   | 3.4   | 4.5   | 12.3  | 18.4  | 23.3  | 27.1  | 28.2  | 26.6  | 24.6  | 19.3  | 12.8  | 6.2  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.5   | 9.3   | 10.3  | 12.1  | 12.0  | 11.7  | 11.8  | 11.8  | 13.0  | 11.8  | 10.0  | 8.2  |
|                                                                                  |             | MCDBR  | 9.7   | 12.1  | 14.2  | 16.5  | 14.5  | 13.6  | 12.8  | 13.0  | 14.3  | 15.1  | 13.8  | 10.1 |
|                                                                                  |             | MCWBR  | 8.3   | 9.5   | 9.6   | 9.7   | 7.2   | 6.3   | 6.3   | 6.7   | 7.0   | 8.6   | 9.9   | 8.7  |
|                                                                                  | 5% WB       | MCDBR  | 9.2   | 11.7  | 12.6  | 14.2  | 12.2  | 11.4  | 11.1  | 11.0  | 11.8  | 12.3  | 12.0  | 9.3  |
|                                                                                  |             | MCWBR  | 8.1   | 9.5   | 9.4   | 9.3   | 7.0   | 6.3   | 6.3   | 6.3   | 6.3   | 8.3   | 9.5   | 8.5  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.291  | 0.308 | 0.341 | 0.395 | 0.422 | 0.430 | 0.443 | 0.432 | 0.405 | 0.351 | 0.308 | 0.291 |      |
|                                                                                  | taud        | 2.419  | 2.379 | 2.389 | 2.251 | 2.228 | 2.247 | 2.206 | 2.231 | 2.280 | 2.430 | 2.500 | 2.443 |      |
|                                                                                  | Ebn at Noon | 861    | 899   | 906   | 874   | 855   | 847   | 831   | 829   | 825   | 838   | 838   | 830   |      |
|                                                                                  | Edh at Noon | 78     | 95    | 107   | 132   | 138   | 136   | 141   | 133   | 118   | 90    | 72    | 70    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.65   | 2.56  | 3.60  | 4.65  | 5.33  | 6.05  | 6.06  | 5.33  | 4.46  | 2.93  | 1.89  | 1.39  |      |
|                                                                                  | RadStd      | 0.18   | 0.33  | 0.29  | 0.38  | 0.42  | 0.44  | 0.34  | 0.31  | 0.38  | 0.34  | 0.17  | 0.15  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (58 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page